

①

Functions of two or more Variables

Definition.

A symbol Z which has a definite value for every pair of values of x and y is called a function of two independent variables x and y and we write $Z = f(x, y)$ or $Z = \phi(x, y)$ or $Z = F(x, y)$.

Example (1).

If $f(x, y) = x^2 + 2y^3$, then $f(3, -1) = (3)^2 + 2(-1)^3 = 7$.

The concept is easily extended. Thus $w = f(x, y, z)$ denotes the value of a function at (x, y, z) [a point in 3 dimensional space R^3], etc.

Domain of a function:

The set values (points) (x, y) for which a function is defined is called the domain of definition or simply domain of the function.

Example (2).

Give the domain of definition for which each of the following functions are defined and real.

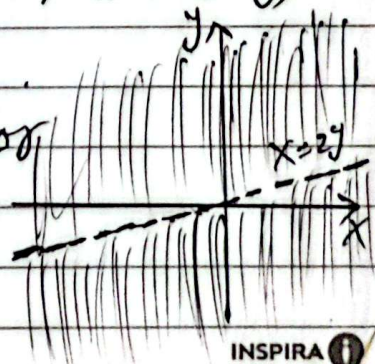
① $Z = 2x^2 + 5xy + y^2$, ③ $Z = \ln(1 - x^2 - y^2)$
② $Z = \frac{x^2 + y^2 - 2xy}{x - 2y}$, ④ $Z = \arcsin(x^2 + y^2 - 3)$.

Solution:

① The domain is defined and real for all points in the XY -plane.

② The Function is defined and real for all points (x, y) such that $x \neq 2y$.

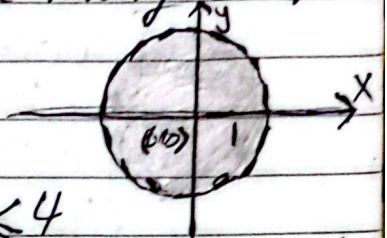
or The Function is defined and real for all points XY -plane except $x = 2y$.



(2)

$$(3) \because 1 - x^2 - y^2 > 0 \Rightarrow x^2 + y^2 < 1$$

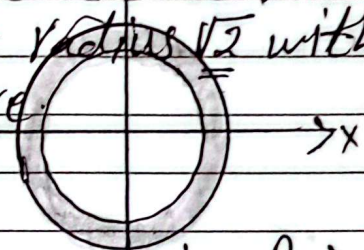
$\therefore$ The domain for which Z is real consists of the set of points (x, y) such that $x^2 + y^2 < 1$, i.e. the set of points inside a circle in the xy -plane having centre at $(0, 0)$ and radius 1.



$$(4) \because Z = \arcsin(x^2 + y^2 - 3) = \sin^{-1}(x^2 + y^2 - 3)$$

$$\therefore -1 \leq x^2 + y^2 - 3 \leq 1 \text{ or } 2 \leq x^2 + y^2 \leq 4$$

$\therefore$ The function is defined and real for all points such that $2 \leq x^2 + y^2 \leq 4$ which is the required domain of definition. This point set consists of all points interior to the circle of radius 2 with centre at the origin and exterior to the circle of radius $\sqrt{2}$ with centre at the origin, as in the figure.



Limits.

The function $f(x, y)$ defined in a region R is said to tend to the limit L as $x \rightarrow a$ and $y \rightarrow b$ if and only if corresponding to a positive number ϵ , there exists another positive number δ such that:

$$|f(x, y) - L| < \epsilon \text{ for } 0 < (x-a)^2 + (y-b)^2 < \delta^2 \text{ for every point } (x, y) \text{ in } R. \text{ i.e. } \lim_{\substack{x \rightarrow a \\ y \rightarrow b}} f(x, y) = L.$$

Example (3).

$$\text{Let } f(x, y) = \begin{cases} 3xy & \text{if } (x, y) \neq (1, 2) \\ 0 & \text{if } (x, y) = (1, 2) \end{cases}$$

As $x \rightarrow 1$ and $y \rightarrow 2$ or $(x, y) \rightarrow (1, 2)$, $f(x, y)$ gets closer to $3(1)(2) = 6$ and we suspect that $\lim_{\substack{x \rightarrow 1 \\ y \rightarrow 2}} f(x, y) = 6$

Note that $\lim_{\substack{x \rightarrow 1 \\ y \rightarrow 2}} f(x, y) \neq f(1, 2)$ since $f(1, 2) = 0$.

(3)

The limit would in fact be $\underline{6}$ even if $f(x,y)$ were not defined at $(1,2)$. Thus the existence of the limit of $f(x,y)$ as $(x,y) \rightarrow (x_0, y_0)$ is in no way dependent on the existence of a value of $f(x,y)$ at (x_0, y_0) .

As in the case of functions of one variable, the following results hold:

If $\lim_{\substack{x \rightarrow a \\ y \rightarrow b}} f(x,y) = L$ and $\lim_{\substack{x \rightarrow a \\ y \rightarrow b}} g(x,y) = m$, then

$$(i) \lim_{\substack{x \rightarrow a \\ y \rightarrow b}} [f(x,y) \pm g(x,y)] = L \pm m,$$

$$(ii) \lim_{\substack{x \rightarrow a \\ y \rightarrow b}} [f(x,y) \cdot g(x,y)] = L \cdot m,$$

$$(iii) \lim_{\substack{x \rightarrow a \\ y \rightarrow b}} [f(x,y)/g(x,y)] = \frac{L}{m}, \text{ where } m \neq 0,$$

$$(iv) \lim_{\substack{x \rightarrow a \\ y \rightarrow b}} [k f(x,y)] = k \lim_{\substack{x \rightarrow a \\ y \rightarrow b}} f(x,y), \text{ } k \text{ is constant,}$$

$$(v) \lim_{\substack{x \rightarrow a \\ y \rightarrow b}} \sqrt{f(x,y)} = \sqrt{\lim_{\substack{x \rightarrow a \\ y \rightarrow b}} f(x,y)}, \text{ where } f(x,y) \geq 0$$

Iterated limits

The iterated limits $\lim_{x \rightarrow x_0} \{ \lim_{y \rightarrow y_0} f(x,y) \}$ and $\lim_{y \rightarrow y_0} \{ \lim_{x \rightarrow x_0} f(x,y) \}$,

[also denoted by $\lim_{x \rightarrow x_0} \lim_{y \rightarrow y_0} f(x,y)$ and $\lim_{y \rightarrow y_0} \lim_{x \rightarrow x_0} f(x,y)$ respectively]

are not necessarily equal. Although they must be equal if $\lim_{\substack{x \rightarrow x_0 \\ y \rightarrow y_0}} f(x,y)$ is to exist, their equality does not

guarantee the existence of this last limit.

(4)

Example (4).

$$\text{If } f(x, y) = \frac{x-y}{x+y}, \text{ then } \lim_{x \rightarrow 0} \left(\lim_{y \rightarrow 0} \frac{x-y}{x+y} \right) = \lim_{x \rightarrow 0} \frac{x}{x} = 1$$

$$\text{and } \lim_{y \rightarrow 0} \left(\lim_{x \rightarrow 0} \frac{x-y}{x+y} \right) = \lim_{y \rightarrow 0} \left(\frac{-y}{y} \right) = \lim_{y \rightarrow 0} (-1) = -1$$

Thus the iterated limits are not equal and so $\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} f(x, y)$ cannot exist.

Continuity

A function $f(x, y)$ is said to be continuous at the point (a, b) if:

- (1) $f(a, b)$ is defined, (2) $\lim_{\substack{x \rightarrow a \\ y \rightarrow b}} f(x, y)$ exists
- (3) $\lim_{\substack{x \rightarrow a \\ y \rightarrow b}} f(x, y) = f(a, b)$.

If a function is continuous at all points of a region then it is said to be continuous in that region.

A function which is not continuous at a point is said to be discontinuous at the point.

If $f(x, y)$, $g(x, y)$ are continuous at (a, b) , then so also are the functions $f(x, y) \pm g(x, y)$, $f(x, y) \cdot g(x, y)$ and $f(x, y)/g(x, y)$ provided $g(x, y) \neq 0$ continuous.

Example (5).

$$\text{Show that the function } f(x, y) = \begin{cases} 3xy, & (x, y) \neq (1, 2) \\ 0, & (x, y) = (1, 2) \end{cases}$$

is discontinuous at $(1, 2)$.

$$\text{Solution. } \because \lim_{\substack{x \rightarrow 1 \\ y \rightarrow 2}} f(x, y) = 3(1)(2) = 6 \text{ and } f(1, 2) = 0$$

$\therefore \lim_{\substack{x \rightarrow 1 \\ y \rightarrow 2}} f(x, y) \neq f(1, 2)$. Hence $f(x, y)$ is not continuous at $(1, 2)$.

(5)

Problems (1)

Evaluate the following limits:

$$\textcircled{1} \lim_{\substack{x \rightarrow 1 \\ y \rightarrow 2}} \frac{2x^2y}{x^2+y^2+1}, \quad \textcircled{2} \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{xy}{x^2+y^2}, \quad \textcircled{3} \lim_{\substack{x \rightarrow 4 \\ y \rightarrow \pi}} \frac{x^2 \sin y}{x}$$

$$\textcircled{4} \lim_{\substack{x \rightarrow 2 \\ y \rightarrow 2}} \frac{xy+1}{x^2+2y^2}, \quad \textcircled{5} \lim_{\substack{x \rightarrow 0^+ \\ y \rightarrow 1^-}} \frac{x+y-1}{\sqrt{x}-\sqrt{1-y}}$$

$$\textcircled{6} \text{ If } f(x,y) = \frac{x-y}{2x+y}, \text{ show that } \lim_{x \rightarrow 0} (\lim_{y \rightarrow 0} f(x,y)) \neq \lim_{y \rightarrow 0} (\lim_{x \rightarrow 0} f(x,y))$$

Also show that the function is discontinuous at the origin.

(7) Investigate the continuity of the function
 $f(x,y) = \begin{cases} xy/(x^2+y^2), & (x,y) \neq (0,0) \\ 0, & (x,y) = (0,0) \end{cases}$
 at the origin.

Solutions

$$\textcircled{1} \lim_{\substack{x \rightarrow 1 \\ y \rightarrow 2}} \frac{2x^2y}{x^2+y^2+1} = \frac{2(1)^2(2)}{(1)^2+(2)^2+1} = \frac{4}{1+4+1} = \frac{4}{6} = \frac{2}{3}$$

$$\textcircled{2} \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{xy}{x^2+y^2}$$

Let $x \rightarrow 0$ and $y \rightarrow 0$ along the line $y = mx$ in the xy -plane

$$\text{Then } \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{xy}{x^2+y^2} = \lim_{x \rightarrow 0} \frac{mx^2}{x^2+m^2x^2} = \lim_{x \rightarrow 0} \frac{mx^2}{x^2(1+m^2)} = \frac{m}{1+m^2}$$

So the limit depends on the approach and therefore does not exist.

or This limit does not exist since along x -axis

$$\lim_{(x,0) \rightarrow (0,0)} \frac{x(0)}{x^2+(0)^2} = 0. \text{ Along the line } x=y \Rightarrow x=y$$

$$\lim_{(0,y) \rightarrow (0,0)} \frac{y^2}{y^2+y^2} = \lim_{y \rightarrow 0} \frac{y^2}{2y^2} = \frac{1}{2}$$

6

Partial derivatives

Let $Z = f(x, y)$ be a function of two variables x and y . If we keep y as constant and vary x alone, then Z is a function of x only. The derivative of Z with respect to x , treating y as constant, is called the partial derivative of Z with respect to x and is denoted by one of the symbols

$$\frac{\partial Z}{\partial x}, \frac{\partial f}{\partial x}, f_x(x, y), D_x f. \text{ Thus } \frac{\partial Z}{\partial x} = \lim_{\delta x \rightarrow 0} \frac{f(x + \delta x, y) - f(x, y)}{\delta x}$$

Similarly, the derivative of Z with respect to y , keeping x as constant, is called the partial derivative of Z with respect to y and is denoted by one of the symbols

$$\frac{\partial Z}{\partial y}, \frac{\partial f}{\partial y}, f_y(x, y), D_y f. \text{ Thus } \frac{\partial Z}{\partial y} = \lim_{\delta y \rightarrow 0} \frac{f(x, y + \delta y) - f(x, y)}{\delta y}$$

Similarly, if Z is a function of three or more variables $x_1, x_2, x_3, \dots$ the partial derivative of Z with respect to x_1 , is obtained by differentiating Z with respect to x_1 , keeping all other variables constant and is written as $\frac{\partial Z}{\partial x_1}$.

In general f_x and f_y are also functions of x and y and so these can be differentiated further partially with respect to x and y .

$$\text{Thus } \frac{\partial}{\partial x} \left(\frac{\partial Z}{\partial x} \right) = \frac{\partial^2 Z}{\partial x^2} \text{ or } \frac{\partial^2 f}{\partial x^2} \text{ or } f_{xx},$$

$$\frac{\partial}{\partial x} \left(\frac{\partial Z}{\partial y} \right) = \frac{\partial^2 Z}{\partial x \partial y} \text{ or } \frac{\partial^2 f}{\partial x \partial y} \text{ or } f_{yx}$$

$$\frac{\partial}{\partial y} \left(\frac{\partial Z}{\partial x} \right) = \frac{\partial^2 Z}{\partial y \partial x} \text{ or } \frac{\partial^2 f}{\partial y \partial x} \text{ or } f_{xy}$$

$$\text{and } \frac{\partial}{\partial y} \left(\frac{\partial Z}{\partial y} \right) = \frac{\partial^2 Z}{\partial y^2} \text{ or } \frac{\partial^2 f}{\partial y^2} \text{ or } f_{yy}.$$

(7)

It can easily be verified that, in all ordinary cases,

$$\frac{\partial^2 Z}{\partial x \partial y} = \frac{\partial^2 Z}{\partial y \partial x}$$

Sometimes we use the following notation

$$\frac{\partial Z}{\partial x} = p, \frac{\partial Z}{\partial y} = q, \frac{\partial^2 Z}{\partial x^2} = r, \frac{\partial^2 Z}{\partial x \partial y} = s, \frac{\partial^2 Z}{\partial y^2} = t.$$

Example (1).

Find the first and second partial derivatives of

$$Z = x^3 + y^3 - 3axy.$$

Sol. We have $Z = x^3 + y^3 - 3axy.$

$$\therefore \frac{\partial Z}{\partial x} = 3x^2 + 0 - 3ay(1) = 3x^2 - 3ay$$

and

$$\frac{\partial Z}{\partial y} = 0 + 3y^2 - 3ax(1) = 3y^2 - 3ax$$

$$\text{Also, } \frac{\partial^2 Z}{\partial x^2} = \frac{\partial}{\partial x} (3x^2 - 3ay) = 6x$$

$$\frac{\partial^2 Z}{\partial y \partial x} = \frac{\partial}{\partial y} (3x^2 - 3ay) = -3a$$

$$\frac{\partial^2 Z}{\partial y^2} = \frac{\partial}{\partial y} (3y^2 - 3ax) = 6y, \text{ and } \frac{\partial^2 Z}{\partial x \partial y} = \frac{\partial}{\partial x} (3y^2 - 3ax) = -3a.$$

$$\text{We observe that } \frac{\partial^2 Z}{\partial y \partial x} = \frac{\partial^2 Z}{\partial x \partial y}.$$

Example (2).

If $u = x^2 \tan^{-1} \frac{y}{x} - y^2 \tan^{-1} \frac{x}{y}$, show that $\frac{\partial^2 u}{\partial x \partial y} = \frac{x^2 - y^2}{x^2 + y^2}$ and $\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial y \partial x}$

Sol. We have $\frac{\partial u}{\partial y} = x^2 \frac{1}{1 + (y/x)^2} \cdot \left(\frac{1}{x}\right) - \left\{ 2y \tan^{-1} \frac{x}{y} + y^2 \frac{1}{1 + (x/y)^2} \left(-\frac{x}{y^2}\right) \right\}$

$$= \frac{x^3}{x^2 + y^2} - 2y \tan^{-1} \frac{x}{y} + \frac{xy^2}{x^2 + y^2} = \frac{x(x^2 + y^2)}{x^2 + y^2} - 2y \tan^{-1} \frac{x}{y}$$

$$\therefore \frac{\partial u}{\partial y} = x - 2y \tan^{-1} \frac{x}{y}.$$

(8)

$$\therefore \frac{\partial^2 u}{\partial x \partial y} = \frac{\partial}{\partial x} \left\{ x - 2y \tan^{-1} \frac{x}{y} \right\} = 1 - 2y \cdot \frac{1}{1+(x/y)^2} \cdot \frac{1}{y} = 1 - \frac{2y^2}{x^2+y^2}$$

$$= \frac{x^2+y^2-2y^2}{x^2+y^2} = \frac{x^2-y^2}{x^2+y^2}$$

$$\text{Similarly, } \frac{\partial u}{\partial x} = 2x \tan^{-1} \frac{y}{x} + x^2 \cdot \frac{1}{1+(y/x)^2} \cdot \left(-\frac{y}{x^2} \right) - y^2 \cdot \frac{1}{1+(x/y)^2} \cdot \left(\frac{1}{y} \right)$$

$$= 2x \tan^{-1} \frac{y}{x} + \left(-\frac{x^2 y}{x^2+y^2} \right) - \frac{y^3}{x^2+y^2}$$

$$= 2x \tan^{-1} \frac{y}{x} - \frac{y(x^2+y^2)}{x^2+y^2} = 2x \tan^{-1} \frac{y}{x} - y$$

and

$$\frac{\partial^2 u}{\partial y \partial x} = \frac{\partial}{\partial y} \left\{ 2x \tan^{-1} \frac{y}{x} - y \right\} = 2x \cdot \frac{1}{1+(y/x)^2} \cdot \frac{1}{x} - 1$$

$$= \frac{2x^2}{x^2+y^2} - 1 = \frac{2x^2 - x^2 - y^2}{x^2+y^2} = \frac{x^2 - y^2}{x^2+y^2}$$

Hence the result.

Example (3).If $Z = f(x+ct) + \phi(x-ct)$, prove that $\frac{\partial^2 Z}{\partial t^2} = c^2 \frac{\partial^2 Z}{\partial x^2}$.

Solution.

$$\text{We have } \frac{\partial Z}{\partial x} = f'(x+ct) \cdot \frac{\partial}{\partial x}(x+ct) + \phi'(x-ct) \cdot \frac{\partial}{\partial x}(x-ct)$$

$$= f'(x+ct) + \phi'(x-ct)$$

$$\text{and } \frac{\partial^2 Z}{\partial x^2} = f''(x+ct) + \phi''(x-ct) \quad \text{--- (i)}$$

$$\text{Again } \frac{\partial Z}{\partial t} = f'(x+ct) \cdot \frac{\partial}{\partial t}(x+ct) + \phi'(x-ct) \cdot \frac{\partial}{\partial t}(x-ct)$$

$$= f'(x+ct)(c) + \phi'(x-ct)(-c)$$

$$= c f'(x+ct) - c \phi'(x-ct)$$

$$\text{and } \frac{\partial^2 Z}{\partial t^2} = \frac{\partial}{\partial t} \{ c f'(x+ct) - c \phi'(x-ct) \} = c^2 f''(x+ct) + c^2 \phi''(x-ct)$$

$$= c^2 [f''(x+ct) + \phi''(x-ct)] \quad \text{--- (ii)}$$

From (i) and (ii), it follows that $\frac{\partial^2 Z}{\partial t^2} = c^2 \frac{\partial^2 Z}{\partial x^2}$.

(9)

Example (4)

If $\theta = t^n e^{-r^2/4t}$, what value of n will make

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \theta}{\partial r} \right) = \frac{\partial \theta}{\partial t} ?$$

Sol. We have $\frac{\partial \theta}{\partial r} = t^n \cdot e^{-r^2/4t} \cdot \left(\frac{-2r}{4t} \right) = -\frac{r}{2} t^{n-1} e^{-r^2/4t}$

$$\therefore r^2 \frac{\partial \theta}{\partial r} = -\frac{r^3}{2} t^{n-1} e^{-r^2/4t}$$

and $\frac{\partial}{\partial r} \left(r^2 \frac{\partial \theta}{\partial r} \right) = -\frac{3r^2}{2} t^{n-1} e^{-r^2/4t} - \frac{r^3}{2} t^{n-1} e^{-r^2/4t} \cdot \left(\frac{-2r}{4t} \right)$

$$= -\frac{3r^2}{2} t^{n-1} e^{-r^2/4t} + \frac{r^4}{4} t^{n-2} e^{-r^2/4t}$$

$$\therefore \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \theta}{\partial r} \right) = \left(-\frac{3}{2} t^{n-1} + \frac{r^2}{4} t^{n-2} \right) e^{-r^2/4t}$$

Also $\frac{\partial \theta}{\partial t} = n t^{n-1} e^{-r^2/4t} + t^n \cdot e^{-r^2/4t} \cdot \frac{r^2}{4t^2}$

$$= \left(n t^{n-1} + \frac{1}{4} r^2 t^{n-2} \right) e^{-r^2/4t}$$

Since $\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \theta}{\partial r} \right) = \frac{\partial \theta}{\partial t}$,

$$\therefore \left(-\frac{3}{2} t^{n-1} + \frac{1}{4} r^2 t^{n-2} \right) e^{-r^2/4t} = \left(n t^{n-1} + \frac{1}{4} r^2 t^{n-2} \right) e^{-r^2/4t}$$

$$\Rightarrow \left(n t^{n-1} + \frac{1}{4} r^2 t^{n-2} + \frac{3}{2} t^{n-1} - \frac{1}{4} r^2 t^{n-2} \right) e^{-r^2/4t} = 0$$

$$\Rightarrow \left(n + \frac{3}{2} \right) t^{n-1} e^{-r^2/4t} \Rightarrow n + \frac{3}{2} = 0$$

Hence

$$n = -\frac{3}{2}$$

Example (5)

If $v = (x^2 + y^2 + z^2)^{-1/2}$, prove that

$$\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} = 0 \quad (\text{Laplace equation})$$

Solution. We have $\frac{\partial v}{\partial x} = -\frac{1}{2} (x^2 + y^2 + z^2)^{-3/2} \cdot 2x = -x (x^2 + y^2 + z^2)^{-3/2}$

(10)

$$\begin{aligned}\frac{\partial^2 v}{\partial x^2} &= - \left[1 \cdot (x^2 + y^2 + z^2)^{-3/2} + x \left(-\frac{3}{2} \right) (x^2 + y^2 + z^2)^{-5/2} \cdot 2x \right] \\ &= - (x^2 + y^2 + z^2)^{-5/2} [x^2 + y^2 + z^2 - 3x^2] \\ &= (x^2 + y^2 + z^2)^{-5/2} (2x^2 - y^2 - z^2).\end{aligned}$$

Similarly,

$$\frac{\partial^2 v}{\partial y^2} = - \frac{1}{2} (x^2 + y^2 + z^2)^{-3/2} \cdot 2y = -y (x^2 + y^2 + z^2)^{-3/2}$$

$$\begin{aligned}\frac{\partial^2 v}{\partial y^2} &= - \left[1 \cdot (x^2 + y^2 + z^2)^{-3/2} + y \left(-\frac{3}{2} \right) (x^2 + y^2 + z^2)^{-5/2} \cdot 2y \right] \\ &= (x^2 + y^2 + z^2)^{-5/2} [- (x^2 + y^2 + z^2) + 3y^2]\end{aligned}$$

$$= (x^2 + y^2 + z^2)^{-5/2} (-x^2 + 2y^2 - z^2)$$

and

$$\frac{\partial^2 v}{\partial z^2} = (x^2 + y^2 + z^2)^{-5/2} (-x^2 - y^2 + 2z^2)$$

$$\begin{aligned}\text{Hence } \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} &= (x^2 + y^2 + z^2)^{-5/2} (2x^2 - y^2 - z^2 - x^2 + 2y^2 - z^2 - x^2 - y^2 + 2z^2) \\ &= (x^2 + y^2 + z^2)^{-5/2} (0) = 0\end{aligned}$$

Obs.

A function v satisfying the Laplace equation is said to be a harmonic function.

Example (6).

If $u = x^y$, show that $\frac{\partial^3 u}{\partial x^2 \partial y} = \frac{\partial^3 u}{\partial x \partial y \partial x}$

Sol. We have $\frac{\partial u}{\partial y} = x^y \log x$

$$\text{and } \frac{\partial^2 u}{\partial x \partial y} = y x^{y-1} \log x + x^y \cdot \frac{1}{x} = x^{y-1} (y \log x + 1)$$

$$\therefore \frac{\partial^3 u}{\partial^2 x \partial y} = \frac{\partial}{\partial x} [x^{y-1} (y \log x + 1)] \quad \dots (i)$$

$$\text{Again } \frac{\partial u}{\partial x} = y x^{y-1} \text{ and } \frac{\partial^2 u}{\partial y \partial x} = 1 \cdot x^{y-1} + y \left(\frac{1}{x} x^y \log x \right) = x^{y-1} (1 + y \log x)$$

$$\therefore \frac{\partial^3 u}{\partial x \partial y \partial x} = \frac{\partial}{\partial x} [x^{y-1} (1 + y \log x)] \quad \dots (ii) \text{ From (i) and (ii) follows the required result}$$

11

Problems (2)

1. Evaluate $\partial Z / \partial x$ and $\partial Z / \partial y$, if

- (i) $Z = x^2y - x \sin xy$; (ii) $Z = \log(x^2 + y^2)$;
(iii) $Z = \tan^{-1}(x^2 + y^2 / (x + y))$; (iv) $x + y + Z = \log Z$.

2. If $Z(x + y) = x^2 + y^2$, show that $(\frac{\partial Z}{\partial x} - \frac{\partial Z}{\partial y})^2 = 4(1 - \frac{\partial Z}{\partial x} \frac{\partial Z}{\partial y})$

3. If $Z = e^{ax+by} f(ax-by)$, Prove that $b \frac{\partial Z}{\partial x} + a \frac{\partial Z}{\partial y} = 2abZ$.

4. Given $u = e^{r \cos \theta} \cos(rs \sin \theta)$, $v = e^{r \cos \theta} \sin(rs \sin \theta)$;
Prove that $\frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta}$ and $\frac{\partial v}{\partial r} = -\frac{1}{r} \frac{\partial u}{\partial \theta}$.

5. Prove that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ if

- (i) $u = \tan^{-1} a \log(x^2 + y^2) + b \tan^{-1}(y/x)$; (ii) $u = \tan^{-1} \left[\frac{2xy}{x^2 - y^2} \right]$.

6. If $Z = \tan(y + ax) - (y - ax)^{3/2}$, show that $\frac{\partial^2 Z}{\partial x^2} = a^2 \frac{\partial^2 Z}{\partial y^2}$.

7. Verify that $f_{xy} = f_{yx}$, when f is equal to

- (i) $\sin^{-1}(y/x)$, (ii) $\log x \tan^{-1}(x^2 + y^2)$

8. If $f(x, y) = (1 - 2xy + y^2)^{-1/2}$, show that

$$\frac{\partial}{\partial x} \left[(1 - x^2) \frac{\partial f}{\partial x} \right] + \frac{\partial}{\partial y} \left[y^2 \frac{\partial f}{\partial y} \right] = 0.$$

9. If $v = \log(x^2 + y^2 + z^2)$, prove that
 $(x^2 + y^2 + z^2) \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} \right) = 2$

10. If $u = e^{xyz}$, Find the value of $\frac{\partial^3 u}{\partial x \partial y \partial z}$.

Multiple Integrals

Double integrals

The definite integral $\int_a^b f(x) dx$ is defined as the limit of the sum $f(x_1)\delta x_1 + f(x_2)\delta x_2 + \dots + f(x_n)\delta x_n$, where $(n \rightarrow \infty)$ and each of the lengths $\delta x_1, \delta x_2, \dots$ tends to Zero.

A double integral is its counterpart in two dimensions.

Consider a function $f(x, y)$ of the independent variables x, y defined at each point in the finite region R of the xy -plane. Divide R into n elementary areas $\delta A_1, \delta A_2, \dots, \delta A_n$. Let (x_r, y_r) be any point within the r th elementary area δA_r . Consider the sum $f(x_1, y_1)\delta A_1 + f(x_2, y_2)\delta A_2 + \dots + f(x_n, y_n)\delta A_n$, i.e. $\sum_{r=1}^n f(x_r, y_r)\delta A_r$.

The limit of this sum, if it exists, as the number of sub-divisions increases indefinitely and area of each sub-division decreases to Zero, is defined as the double integral of $f(x, y)$ over the region R and is written as $\iint_R f(x, y) dA$.

$$\text{Thus } \iint_R f(x, y) dA = \lim_{\substack{n \rightarrow \infty \\ \delta A \rightarrow 0}} \sum_{r=1}^n f(x_r, y_r) \delta A_r \quad \text{--- (1)}$$

The utility of double integrals would be limited if it were required to take limit of sums to evaluate them. However, there is another method of evaluating double integrals by successive single integrations.

For purposes of evaluation, (1) is expressed as the repeated integral $\int_{x_1}^{x_2} \int_{y_1}^{y_2} f(x, y) dx dy$.

Its value is found as follows:

(13)

(c) When y_1, y_2 are functions of x and x_1, x_2 are constants. $f(x, y)$ is first integrated w.r.t. y keeping x fixed between limits y_1, y_2 and then the resulting expression is integrated w.r.t. x within the limits x_1, x_2 i.e.

$$I_1 = \int_{x_1}^{x_2} \left(\int_{y_1}^{y_2} f(x, y) dy \right) dx$$

where integration is carried from the inner to the outer rectangle.

Fig. (a) illustrates this process. Here AB and CD are the two curves whose equations are $y_1 = f_1(x)$ and $y_2 = f_2(x)$. PQ is a vertical strip of width dx .

Then the inner rectangle integral means that the integration is along one edge of the strip PQ from P to Q (x remaining constant), while the outer rectangle integral corresponds to the sliding of the edge from AC to BD .

Thus the whole region of integration is the area $ABDC$.

(ii) When x_1, x_2 are functions of y and y_1, y_2 are constants, $f(x, y)$ is first integrated w.r.t. x , keeping y fixed, within the limits x_1, x_2 and the resulting expression is integrated w.r.t. y between the limits y_1, y_2 i.e.

$$I_2 = \int_{y_1}^{y_2} \left(\int_{x_1}^{x_2} f(x, y) dx \right) dy$$

which is geometrically illustrated by Fig. (b). Here AB and CD are the curves $x_1 = f_1(y)$ and $x_2 = f_2(y)$. PQ is a horizontal strip of width dy .

Then inner rectangle indicates that the integration is along one edge of this strip from P to Q while the outer rectangle

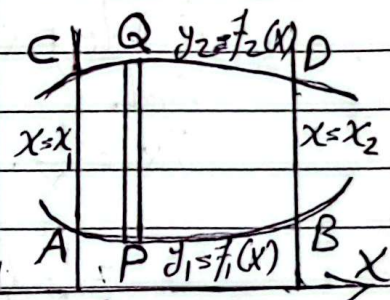


Fig. (a)

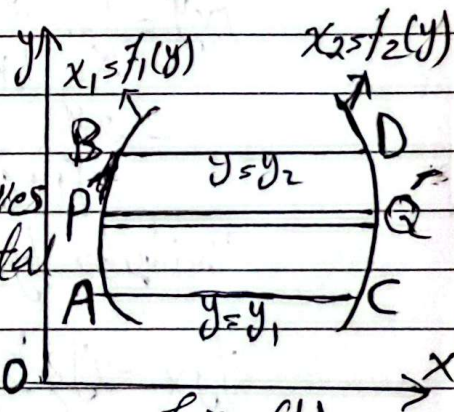


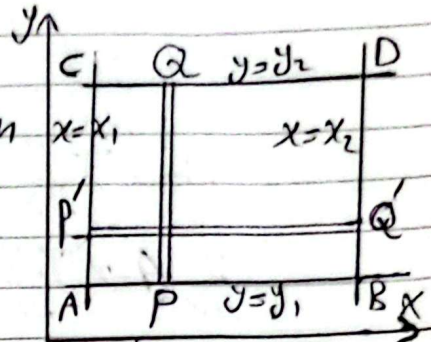
Fig. (b)

(14)

corresponds to the sliding of this edge from AC to BD. Thus the whole region of integration is the area ABDC.

(iii) When both pairs of limits are constants, the region of integration is the rectangle ABDC (fig. C).

In I_1 , we integrate along the vertical strip PQ and then slide it from AC to BD.



In I_2 , we integrate along the horizontal strip $P'Q'$ and then slide it from AB to CD.

Here obviously $I_1 = I_2$.

Thus for constant limits, it hardly matters whether we first integrate w.r.t. x and then w.r.t. y or vice versa.

Example (1).

Evaluate the iterated integral $\int_3^5 \int_{-x}^{x^2} (4x + 10y) dy dx$.

$$\begin{aligned}
 \text{Sol. } \int_3^5 \int_{-x}^{x^2} (4x + 10y) dy dx &= \int_3^5 \left(\int_{-x}^{x^2} (4x + 10y) dy \right) dx \\
 &= \int_3^5 [4xy + 5y^2]_{-x}^{x^2} dx \\
 &= \int_3^5 (4x^3 + 5x^4 - (-4x^2 + 5x^2)) dx \\
 &= \int_3^5 (5x^4 + 4x^3 - x^2) dx \\
 &= \left[x^5 + x^4 - \frac{1}{3}x^3 \right]_3^5 \\
 &= \left[(5)^5 + (5)^4 - \frac{1}{3}(5)^3 - \left((3)^5 + (3)^4 - \frac{1}{3}(3)^3 \right) \right] \\
 &= 3393 \frac{1}{3} \text{ nearly.}
 \end{aligned}$$

(15)

Example (2).Evaluate the iterated integral $\int_0^1 \int_0^{y^2} 2ye^x dx dy$.**Solution:**

$$\begin{aligned}
 \int_0^1 \int_0^{y^2} 2ye^x dx dy &= \int_0^1 \left(\int_0^{y^2} 2ye^x dx \right) dy = \int_0^1 [2ye^x]_0^{y^2} dy \\
 &= 2 \int_0^1 [ye^{y^2} - y] dy = 2 \left\{ \int_0^1 ye^{y^2} dy - \int_0^1 y dy \right\} \\
 &= 2 \left\{ \frac{1}{2} \int_0^1 e^{y^2} dy^2 - \int_0^1 y dy \right\} \\
 &= 2 \left\{ \left(\frac{1}{2} \right) e^{y^2} - \frac{1}{2} y^2 \right\}_0^1 = [e^{y^2} - y^2]_0^1 \\
 &= (e - 1) - (1 - 0) = e - 1 - 1 = e - 2
 \end{aligned}$$

Example (3).Evaluate $\int_0^5 \int_0^{x^2} x(x^2 + y^2) dx dy$ **Solution:**

$$\begin{aligned}
 \int_0^5 \int_0^{x^2} x(x^2 + y^2) dx dy &= \int_0^5 \int_0^{x^2} (x^3 + xy^2) dx dy \\
 &= \int_0^5 \int_0^{x^2} (x^3 + xy^2) dy dx = \int_0^5 \left[x^3 y + \frac{xy^3}{3} \right]_0^{x^2} dx \\
 &= \int_0^5 \left[x^3 \cdot x^2 + x \cdot \frac{x^6}{3} \right] dx = \int_0^5 \left[x^5 + \frac{x^7}{3} \right] dx \\
 &= \left[\frac{x^6}{6} + \frac{x^8}{24} \right]_0^5 = \left[\frac{(5)^6}{6} + \frac{(5)^8}{24} - (0) \right] = 5 \left[\frac{5^5}{6} + \frac{5^7}{24} \right] \\
 &= 18880.2 \text{ nearly.}
 \end{aligned}$$

Example (4). Evaluate $\iint_A xy dx dy$, where A is the domain bounded by x -axis, ordinate $x = 2a$ and the curve $x^2 = 4ay$.

(16)

Solution.

The line $x=2a$ and the parabola $x^2=4ay$ intersect at $L(2a, a)$ $\because x^2=4ay \Rightarrow (2a)^2=4ay \Rightarrow y=a$.

Fig. (1) shows the domain A which is the area OML .

Integrating first over a vertical strip PQ , i.e. w.r.t. y from $P(y=0)$ to $Q(y=x^2/4a)$ on the parabola and then w.r.t. x from $x=0$ to $x=2a$,

$$\iint_A xy \, dx \, dy = \int_0^{2a} \left(\int_0^{x^2/4a} xy \, dy \right) dx$$

$$= \int_0^{2a} x \left[\frac{y^2}{2} \right]_0^{x^2/4a} dx = \int_0^{2a} x \frac{(x^2/4a)^2}{2} dx$$

$$= \int_0^{2a} \frac{x^5}{32a^2} dx = \frac{1}{32a^2} \left[\frac{x^6}{6} \right]_0^{2a} = \frac{1}{32a^2} \frac{(2a)^6}{6}$$

$$= \frac{64a^6}{(32)(6)a^2} = \frac{a^4}{3}$$

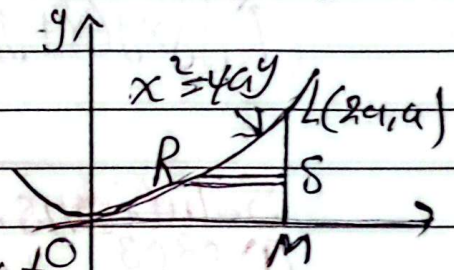
Otherwise integrating first over a horizontal strip RS , i.e. w.r.t. x from $R(x=2\sqrt{ay})$ on the parabola to $S(x=2a)$ and then w.r.t. y from $y=0$ to $y=a$, we get

$$\iint_A xy \, dx \, dy = \int_0^a \int_{2\sqrt{ay}}^{2a} xy \, dx \, dy = \int_0^a y \left[\frac{x^2}{2} \right]_{2\sqrt{ay}}^{2a} dy$$

$$= \int_0^a y \left(\frac{4a^2}{2} - \frac{4ay}{2} \right) dy = 2a \int_0^a (ay - y^2) dy$$

$$= 2a \left[\frac{ay^2}{2} - \frac{y^3}{3} \right]_0^a = 2a \left[\frac{a^3}{2} - \frac{a^3}{3} \right]$$

$$= 2a \left[\frac{3a^3 - 2a^3}{6} \right] = \frac{2a^4}{6} = \frac{a^4}{3}$$



(17)

Problems (3).

Evaluate the following integrals.

$$\textcircled{1} \int_1^2 \int_1^3 x y^2 dx dy, \quad \textcircled{2} \int_0^1 \int_x^{\sqrt{x}} (x^2 + y^2) dx dy$$

$$\textcircled{3} \int_0^4 \int_0^{x^2} e^{y/x} dy dx, \quad \textcircled{4} \int_1^5 \int_0^x \frac{3}{x^2 + y^2} dy dx$$

$$\textcircled{5} \int_{1/2}^1 \int_0^{2x} \cos(\pi x^2) dy dx, \quad \textcircled{6} \int_0^\pi \int_0^\pi \sin x \cdot \sin y dx dy$$

$$\textcircled{7} \iint xy dx dy \text{ over the positive quadrant of the circle } x^2 + y^2 = a^2.$$

$$\textcircled{8} \iint (x+y)^2 dx dy \text{ over the area bounded by the ellipse } x^2/a^2 + y^2/b^2 = 1.$$

$$\textcircled{9} \iint xy(x+y) dx dy \text{ over the area between } y = x^2 \text{ and } y = x.$$

Solutions.

$$\begin{aligned} \textcircled{1} \int_1^2 \int_1^3 x y^2 dx dy &= \int_1^2 \left[\frac{x^2 y^2}{2} \right]_1^3 dy = \int_1^2 \left(\frac{9}{2} - \frac{1}{2} \right) y^2 dy \\ &= \int_1^2 4 y^2 dy = \left[\frac{4}{3} y^3 \right]_1^2 = \frac{4}{3} [8 - 1] = \frac{28}{3} \end{aligned}$$

$$\begin{aligned} \underline{\text{or}} \int_1^2 \int_1^3 x y^2 dx dy &= \int_1^3 \int_1^2 x y^2 dy dx = \int_1^3 \left[x \frac{y^3}{3} \right]_1^2 dx \\ &= \int_1^3 \left(\frac{8}{3} - \frac{1}{3} \right) x dx = \int_1^3 \frac{7}{3} x dx \\ &= \frac{7}{3} \left[\frac{x^2}{2} \right]_1^3 = \frac{7}{3} \left[\frac{9}{2} - \frac{1}{2} \right] = \frac{7}{3} \times 4 = \frac{28}{3} \end{aligned}$$

(18)

Triple Integrals

Consider a function $f(x, y, z)$ defined at every point of the 3-dimensional finite region V . Divide V into n elementary volumes $\delta V_1, \delta V_2, \dots, \delta V_n$. Let (x_r, y_r, z_r) be any point within the r th sub-division δV_r . Consider the sum

$$\sum_{r=1}^n f(x_r, y_r, z_r) \delta V_r.$$

The limit of this sum, if it exists, as $n \rightarrow \infty$ and $\delta V_r \rightarrow 0$ is called the triple integral of $f(x, y, z)$ over the region V and is denoted by $\iiint_V f(x, y, z) dV$.

For purposes of evaluation, it can also be expressed as the repeated integral $\int_{x_1}^{x_2} \int_{y_1}^{y_2} \int_{z_1}^{z_2} f(x, y, z) dx dy dz$.

If x_1, x_2 are constants; y_1, y_2 are either constants or functions of x and z_1, z_2 are either constants or functions of x and y ; then this integral is evaluated as follows:

First $f(x, y, z)$ is integrated w.r.t. z between the limits z_1 and z_2 keeping x and y fixed. The resulting expression is integrated w.r.t. y between the limits y_1 and y_2 keeping x constant. The result just obtained is finally integrated w.r.t. x from x_1 to x_2 .

$$\text{Thus } I = \int_{x_1}^{x_2} \left[\int_{y_1(x)}^{y_2(x)} \left[\int_{z_1(x, y)}^{z_2(x, y)} f(x, y, z) dz \right] dy \right] dx,$$

where the integration is carried out from the innermost ~~rectangle~~ brackets to the outermost brackets.

The order of integration may be different for different types of limits.

(19)

Example (1) Evaluate $\int_{-1}^1 \int_0^z \int_{x-z}^{x+z} (x+y+z) dx dy dz$

Solution. Integrating first w.r.t. y keeping x and z const. we have

$$\begin{aligned} I &= \int_{-1}^1 \int_0^z \int_{x-z}^{x+z} (x+y+z) dy dx dz = \int_{-1}^1 \int_0^z \left[xy + \frac{y^2}{2} + yz \right]_{x-z}^{x+z} dx dz \\ &= \int_{-1}^1 \int_0^z \left[x(x+z) + \frac{(x+z)^2}{2} + (x+z)z - x(x-z) - \frac{(x-z)^2}{2} - (x-z)z \right] dx dz \\ &= \int_{-1}^1 \int_0^z \left[x^2 + xz + \frac{x^2}{2} + xz + \frac{z^2}{2} + xz + z^2 - x^2 + \frac{x^2}{2} + xz - \frac{z^2}{2} - xz + z^2 \right] dx dz \\ &= \int_{-1}^1 \int_0^z (4xz + 2z^2) dx dz = \int_{-1}^1 [2x^2z + 2xz^2]_0^z dz \\ &= \int_{-1}^1 [2z^3 + 2z^3] dz = \int_{-1}^1 (4z^3) dz = \left[z^4 \right]_{-1}^1 \\ &= (1)^4 - (-1)^4 = 1 - 1 = 0 \end{aligned}$$

Example (2) Evaluate $\int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} xyz dx dy dz$

Solution.

$$\begin{aligned} I &= \int_0^1 x \left[\int_0^{\sqrt{1-x^2}} y \left\{ \int_0^{\sqrt{1-x^2-y^2}} z dz \right\} dy \right] dx \\ &= \int_0^1 x \left\{ \int_0^{\sqrt{1-x^2}} y \left[\frac{z^2}{2} \right]_0^{\sqrt{1-x^2-y^2}} dy \right\} dx \\ &= \int_0^1 x \left\{ \int_0^{\sqrt{1-x^2}} y \cdot \frac{1}{2} (1-x^2-y^2) dy \right\} dx = \int_0^1 x \left\{ \frac{1}{2} \int_0^{\sqrt{1-x^2}} ((1-x^2)y - y^3) dy \right\} dx \\ &= \frac{1}{2} \int_0^1 x \left[\frac{(1-x^2)y^2}{2} - \frac{y^4}{4} \right]_0^{\sqrt{1-x^2}} dx = \frac{1}{8} \int_0^1 \{ (1-x^2)^2 \cdot 2x - (1-x^2)^4 \cdot x \} dx \\ &= \frac{1}{8} \int_0^1 [2x - 4x^3 + 2x^5 - (1 - 4x^2 + 6x^4 - 4x^6 + x^8)] dx \\ &= \frac{1}{8} \int_0^1 x(1-x^2)^2 dx = \frac{1}{8} \int_0^1 (x - 2x^3 + x^5) dx = \frac{1}{8} \left[\frac{x^2}{2} - \frac{2x^4}{4} + \frac{x^6}{6} \right]_0^1 \\ &= \frac{1}{8} \left(\frac{1}{2} - \frac{1}{2} + \frac{1}{6} \right) = \frac{1}{48} \end{aligned}$$

20

Problems (4)

Evaluate the following integrals:

① $\int_0^1 \int_0^2 \int_1^2 x^2 y z \, dx \, dy \, dz$

② $\int_{-c}^c \int_{-b}^b \int_{-a}^a (x^2 + y^2 + z^2) \, dx \, dy \, dz$

③ $\int_0^4 \int_0^{2\sqrt{z}} \int_0^{\sqrt{4z-x^2}} dy \, dx \, dz$

④ $\int_0^a \int_0^x \int_0^{x+y} e^{x+y+z} \, dz \, dy \, dx$

⑤ $\int_1^e \int_1^{\log y} \int_1^{e^x} \log z \, dz \, dx \, dy$

⑥ $\int_0^{\pi/2} \int_0^{a \sin \theta} \int_0^{\frac{a^2-r^2}{a}} r \, dz \, dr \, d\theta$

Solutions.

① $\int_0^1 \int_0^2 \int_1^2 x^2 y z \, dx \, dy \, dz = \int_0^1 x^2 \int_0^2 y \int_1^2 z \, dz \, dy \, dx$
 $= \int_0^1 x^2 \int_0^2 y \left[\frac{z^2}{2} \right]_1^2 dy \, dx$
 $= \int_0^1 x^2 \int_0^2 y \left[\frac{4}{2} - \frac{1}{2} \right] dy \, dx = \frac{3}{2} \int_0^1 x^2 \left[\frac{y^2}{2} \right]_0^2 dx$
 $= \frac{3}{2} \int_0^1 x^2 \left[\frac{4}{2} - 0 \right] dx = 3 \int_0^1 x^2 \, dx = 3 \left[\frac{x^3}{3} \right]_0^1$
 $= 3 \left[\frac{1}{3} - 0 \right] = 1$

② $\int_{-c}^c \int_{-b}^b \int_{-a}^a (x^2 + y^2 + z^2) \, dx \, dy \, dz = \int_{-c}^c \int_{-b}^b \left[\frac{x^3}{3} + xy^2 + xz^2 \right]_{-a}^a dy \, dz$
 $= \int_{-c}^c \int_{-b}^b \left(\frac{a^3}{3} + ay^2 + az^3 + \frac{a^3}{3} + ay^2 + az^3 \right) dy \, dz$